

AJOY KUMAR SARKER

Dhaka, Bangladesh
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Education

Bangladesh University of Engineering & Technology Apr 2019-Jun 2024
Bachelor of Science in Mechanical Engineering
CGPA: 3.42 out of 4.00

Related Coursework

Solid Mechanics, Composite Materials, Production Processes, Metallic Materials, Numerical Analysis, Mechanics of Machinery, Polymer Chemistry, bio-engineering

Standardized Test Score

International English Language Testing System (IELTS) Nov 15, 2024
Overall Score: 7.5/9.0 (Listening: 7.5, Reading: 9, Speaking: 6, Writing: 6.5)

Research Interest

Additive Manufacturing, Composite Materials & Ceramics, Biomaterials, Fracture Mechanics, Molecular Dynamics, Computational Mechanics, Soft Robotics, Machine Learning, Biomechanics

Research Experience

Undergraduate Thesis | Supervisor: Dr. Md. Afsar Ali

Software & Programming Language: LAMMPS, Python, C, OVITO

- **Thesis Title:** Analysis of Sintering Behavior and Thermo-Mechanical Properties of Tungsten-Copper Functionally Graded Materials: A Molecular Dynamics Study on Spark Plasma Sintering
- Manually created potential file generated from DFT parameters
- Performed Molecular Dynamics Simulations
- Explored and analyzed different thermal and mechanical properties of the FGM parts after Sintering
- Analyzed the influence of the constituent particles' size during fabrication on the sintering Behavior

Performance Analysis of a Newly Developed Soft Robotics Fishtail (running)

- Final version of the undergraduate project. Manufactured a new robotic fishtail using silicon
- Developed a system to conduct test
- Analysis performances on different conditions e.g. angle of attack, flapping velocity, motor Frequency

Conference Proceeding

Muhit M Abrar, **Kumar Sarker Ajoy**, A. M. Afsar, "Analysis of Sintering Behavior and Thermal Properties of Tungsten-Copper Functionally Graded Materials: A Molecular Dynamics Study on Spark Plasma Sintering", *9th BSME International Conference on Thermal Engineering (BSME ICTE), 2024*

Projects

Optimization of the nozzle control of traditional cooling tower

Mar 2023

Developed a new nozzle with a feedback system that can change the flow of water needed instead of using a conventional nozzle. An Excel-based add-on was used to perform all necessary calculations for cooling load. A motor closed/opened the holes of the nozzle and controlled the flow.

- Supervisor: Dr. A K M Monjur Morshed
- Objective: Reduce the wastage of fresh water.
- Software: SolidWorks, MS Excel, HTRI
- Manufacturing Techniques: CNC Machining

Hilsha bot-A bio-inspired fish

Sept 2022

Consists of three subsystems-electric system, a Buoyancy Control Unit (BCU), soft actuator. Fish could control its depth using vacuum-driven BCU without changing overall weight. The actuator was manufactured using locally available silicon through injection molding.

- Supervisor: Dr. Kazi Arafat Rahman
- Objective: Underwater deep-sea exploration and surveillance
- Software & Programming language: SolidWorks, Altium Pro, Fusion 360, C, Python
- Manufacturing Techniques & Automation tools: Injection molding, Laser Cutting, 3D Printing.
- Video Demonstration:

https://www.youtube.com/watch?v=Rzq9v2fQp0&ab_channel=AjoyKumarSarker

Technical Experience

Body structure Sub Team Leader | Team Interplanetary

Oct 2022-Dec 2023

- Design & Development of a new suspension system for better modularity
- Represented the team in person at European Rover Challenge (ERC) 2022 onsite in, Poland.

SolidWorks Instructor | BUET Automobile Club

Dec 2022-Mar 2023

- Collaborated with a team of instructors to develop a course curriculum
- Personally delivered lectures on SOLIDWORKS, focusing on efficient 3D modeling and solving complex modeling challenges

Industrial Trainee | Ashuganj Power Station Company Limited (APSCL), Ashuganj, Bangladesh

- 21 days long industrial training.
- Insightful discussion with professionals about the current energy scenario in Bangladesh.
- Hands-on experience in problem-finding methods and mechanical maintenance of gas turbine.
- Assigned department: Mechanical Maintenance, Control room

Achievements

- 8th position worldwide at European Rover Challenge 2022, Kielce, Poland
- 4th Position Globally - European Rover Challenge 2022 Remote, Kielce, Poland
- Vice Chancellor Award for European Rover Challenge 2022
- Vice Chancellor Award for International Planetary Aerial System Challenge -2021
- Innovation Award & 8th (Globally) at International Planetary Aerial System Challenge-2021
- Final Participant, IIUC National Photography Competition

Skills & Certifications

- **LAMMPS:** Proficient in LAMMPS coding and OVITO simulation for microstructural analysis.
- **MATLAB:** Certified by MathWorks. Completed 4.5 credit courses with A+ offered by the Department of Mechanical Engineering, BUET.
- **Finite Element Analysis:** Abaqus
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, Seaborn
- **ML Framework:** Scikit-learn
- **CAD software:** AutoCAD & SolidWorks.
- **Programming Language:** Experienced in writing codes using C and Python.

On-campus Activities

General Secretary	Joint Secretary	Association Representative
BUET Film Society	BUET Automobile Club	Mechanical Engineering Association, BUET
January 2023-August 2024	July 2023-September 2024	February 2020-December 2022